Environmental Telemetry System

An automated measurement and remote transmission system for environmental data including temperature, humidity, air quality, and light intensity.

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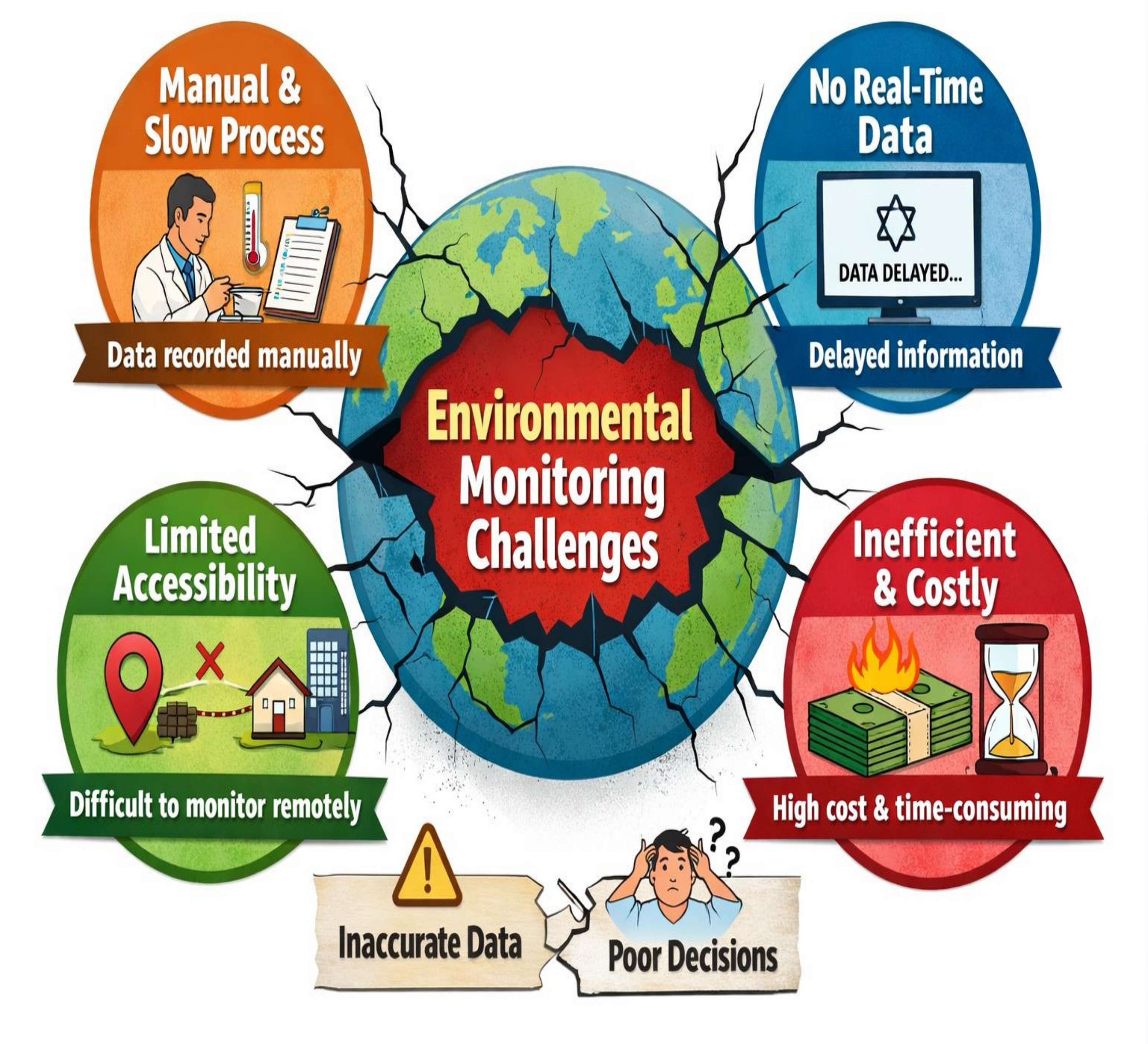
Problem statement



Environmental Monitoring Challenges

Traditional manual data collection methods are time-consuming, error-prone, and lack real-time capabilities. Industries, agriculture, and smart cities require continuous, remote monitoring of environmental conditions.

Key Points

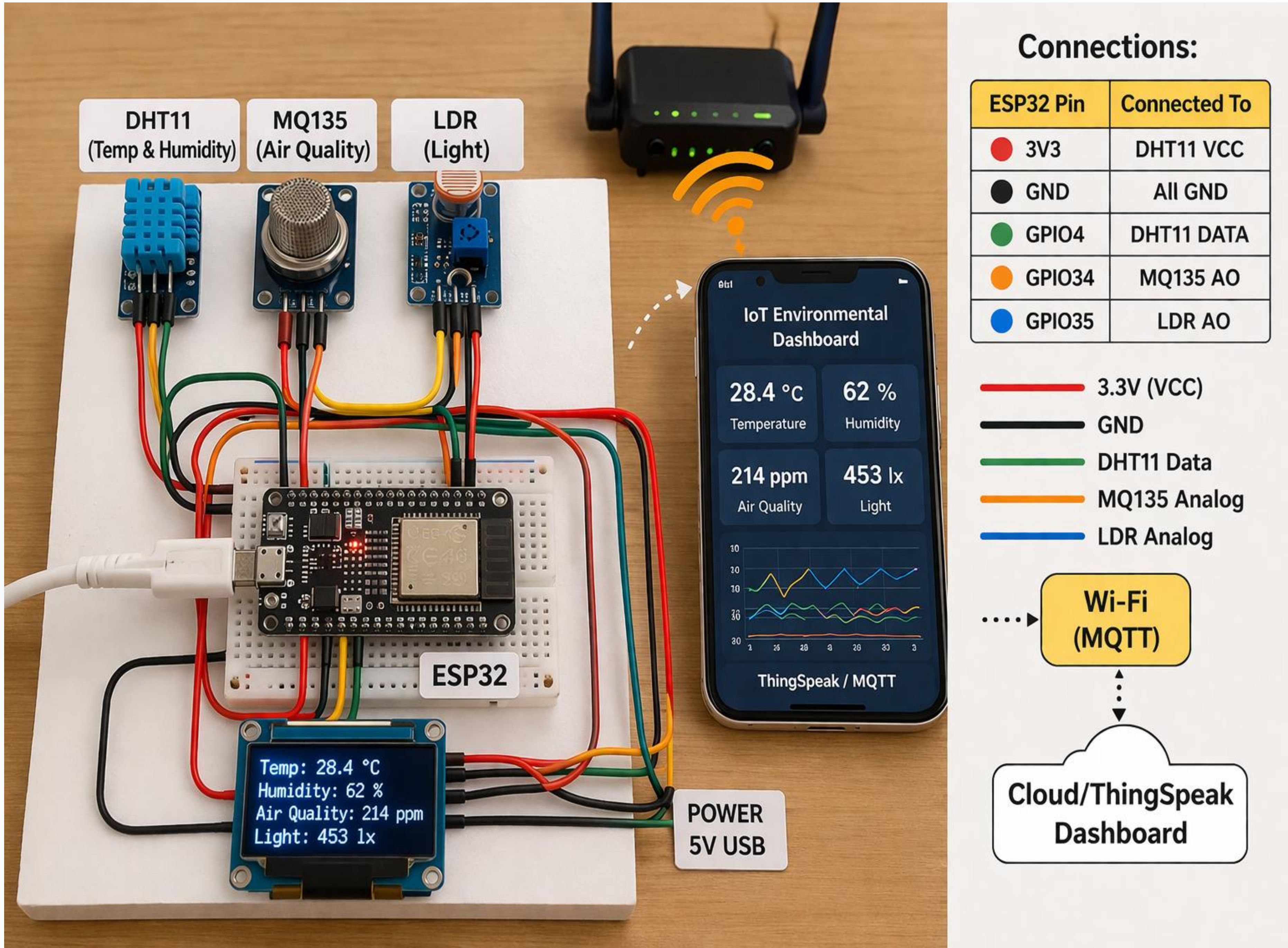
- Delayed response to environmental changes.
- High labour costs for manual readings
- Limited accessibility to remote locations
- Inability to track trends over time

Solution

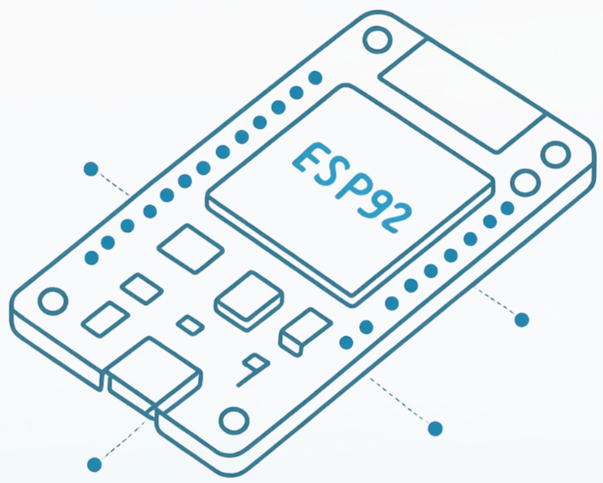
1.Automatic Real-Time Monitoring
The system continuously collects environmental data (temperature, humidity, air quality, light) using sensors and updates it in real time.

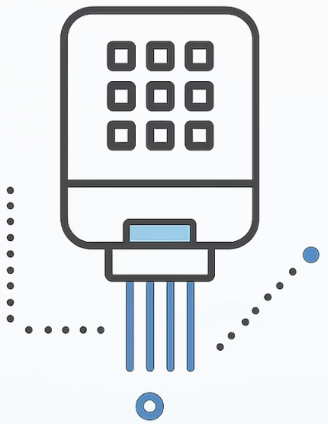
2.Wireless Data Transmission using IoT
Data is sent through the internet using MQTT protocol, allowing fast and reliable communication between the ESP32 and cloud platforms.

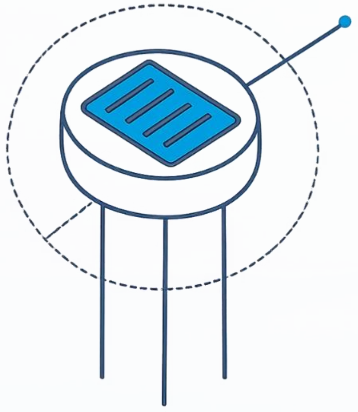
3.Remote Access and Alerts
Users can monitor data from anywhere using mobile or computer, and receive alerts when environmental conditions become unsafe.



Hardware Components







ESP32 Board

Main controller with built-in Wi-Fi and Bluetooth for wireless communication

DHT22 Sensor

Measures temperature and humidity with high accuracy

MQ135 Sensor

Detects air quality and harmful gases in the environment

LDR Sensor

Measures light intensity using photoresistive properties

Additional components include breadboard, jumper wires, and USB power supply for complete circuit assembly.

Software Stack

MQTT

Lightweight protocol using publish-subscribe model sends data from ESP32 to server

Wi-Fi

Connects ESP32 to internet. Enables remote monitoring

Security

Protects data. Prevents unauthorised access

Arduino IDE

Primary programming environment for writing and uploading code to ESP32 microcontroller

MQTT Broker

HiveMQ or Mosquitto handles lightweight data transmission between ESP32 and server

Dashboard Platforms

Node-RED, ThingSpeak, or Blynk for real-time data visualization and monitoring

System Architecture & Working Principle

Sensors

Measure environmental parameters

ESP32

Processes and formats data

Wi-Fi

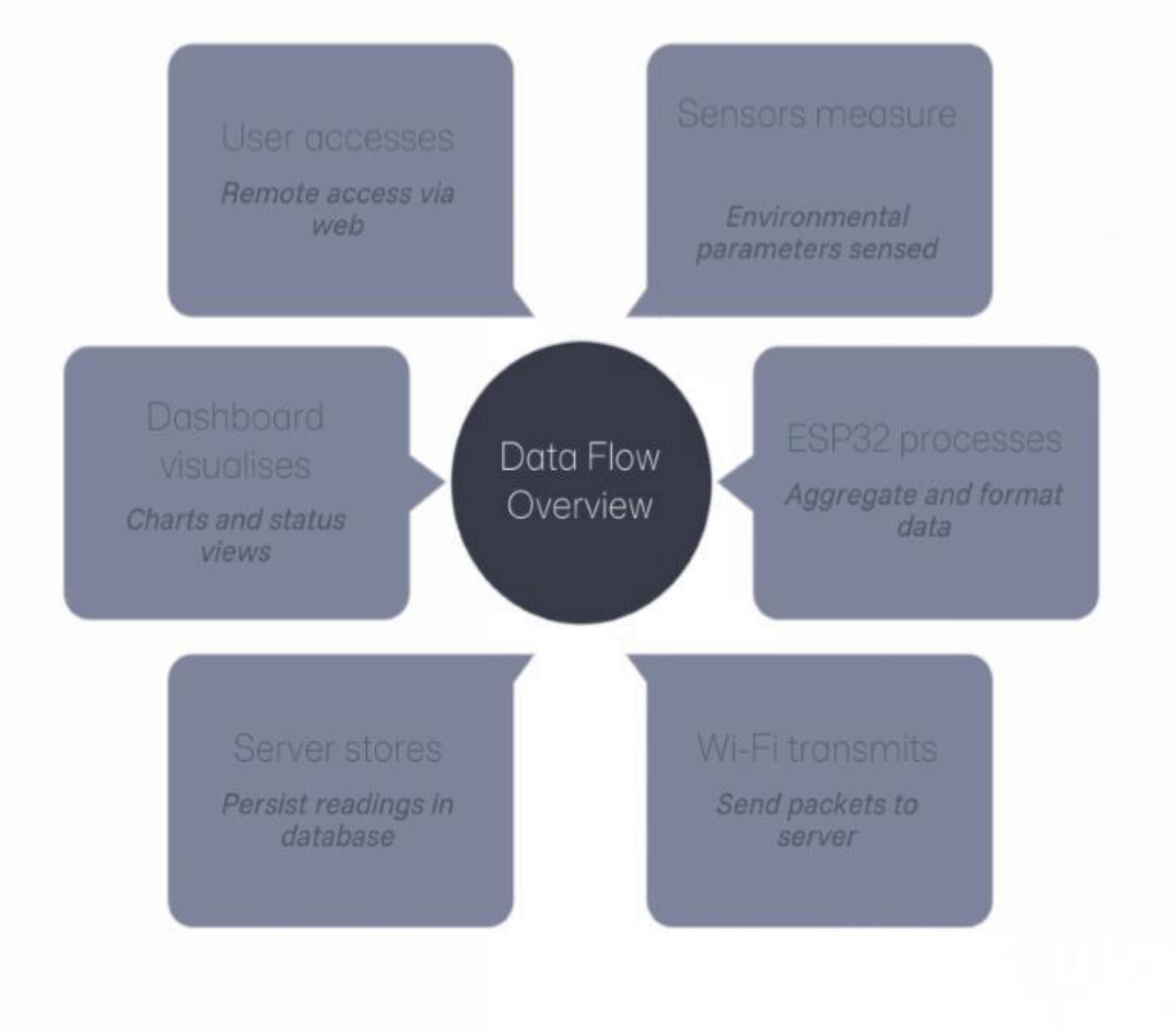
Wireless transmission

Server

Receives and stores data

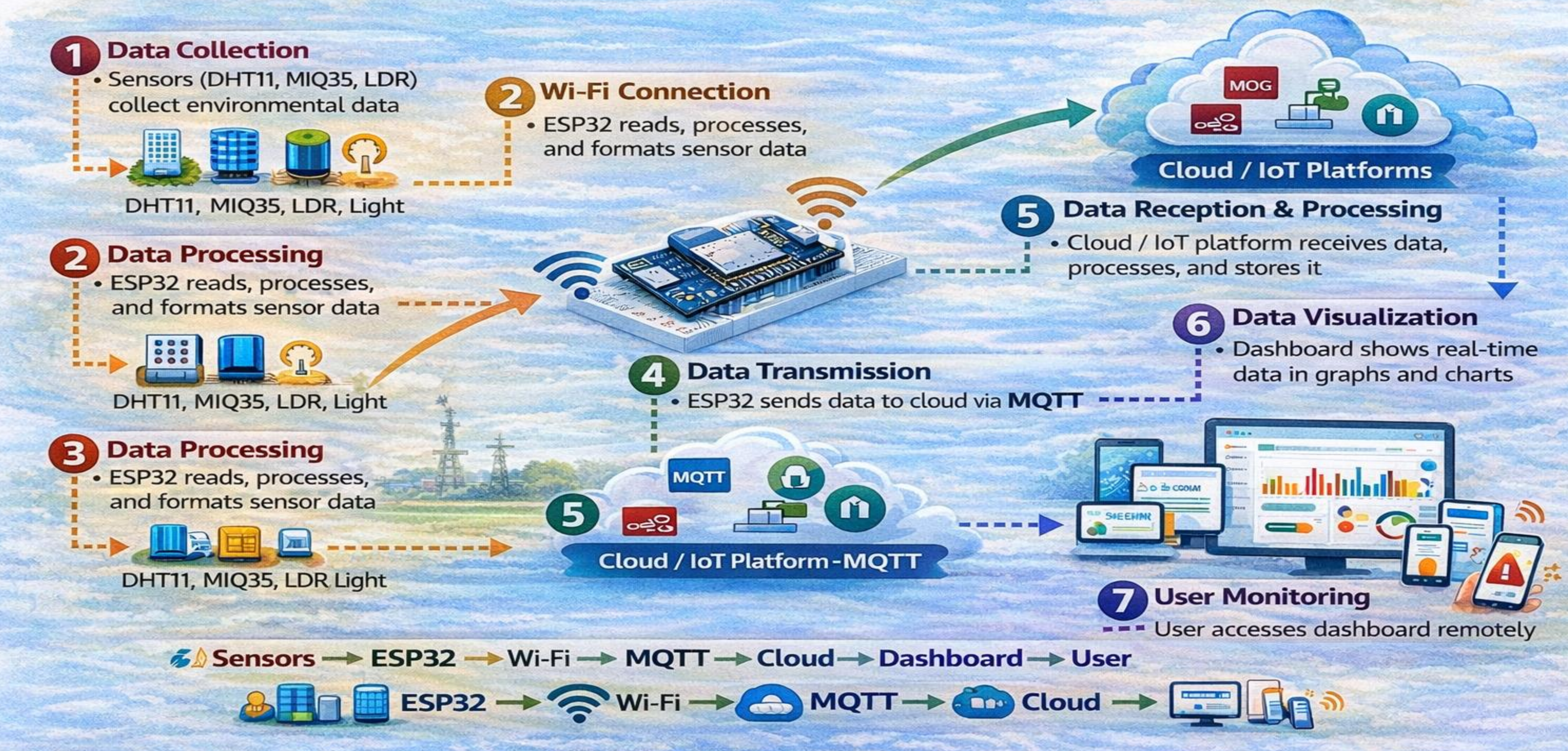
Dashboard

Visualizes real-time data



The system operates continuously: sensors detect parameters, ESP32 reads through GPIO pins, connects to Wi-Fi, transmits via MQTT protocol, and displays on dashboard in graphs and gauges.

Working of the system



Circuit Connections

Power Setup (Common for All)

Arduino 5V → Breadboard + rail

Arduino GND → Breadboard – rail

All components use this power

2. Temperature Sensor (TMP36)

Flat side facing you

Left (VCC) → 5V

Middle (Vout) → A2

Right (GND) → GND

3. Gas Sensor (6-Pin MQ Sensor)

Heater:

H1 → 5V

H2 → GND

Signal:

Any A pin → 5V

Any B pin → A0

Important:

10kΩ resistor between A0 and GND

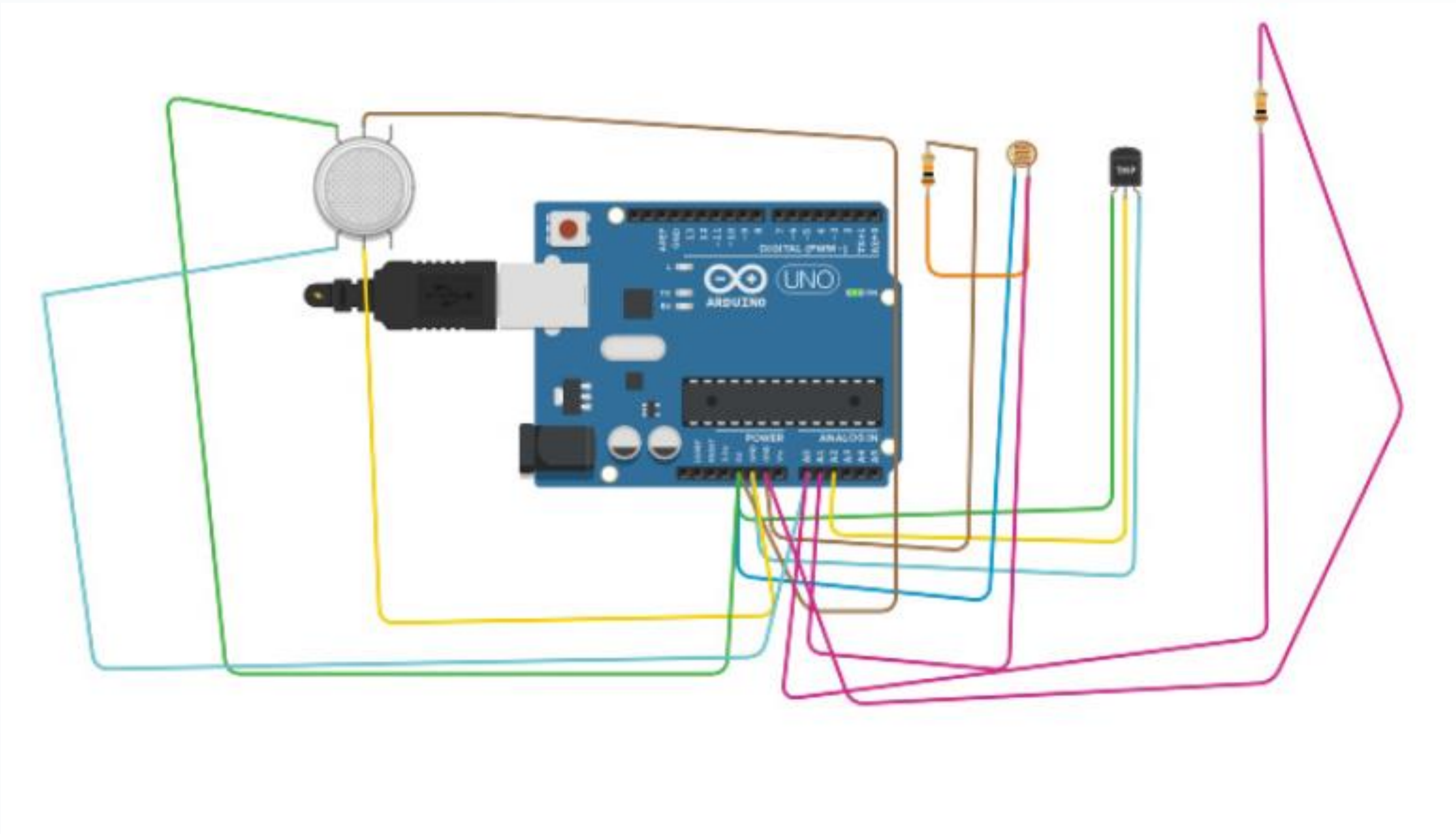
4. LDR (Photoresistor)

Use voltage divider

One side → 5V

Other side → A1

10kΩ resistor from A1 → GND

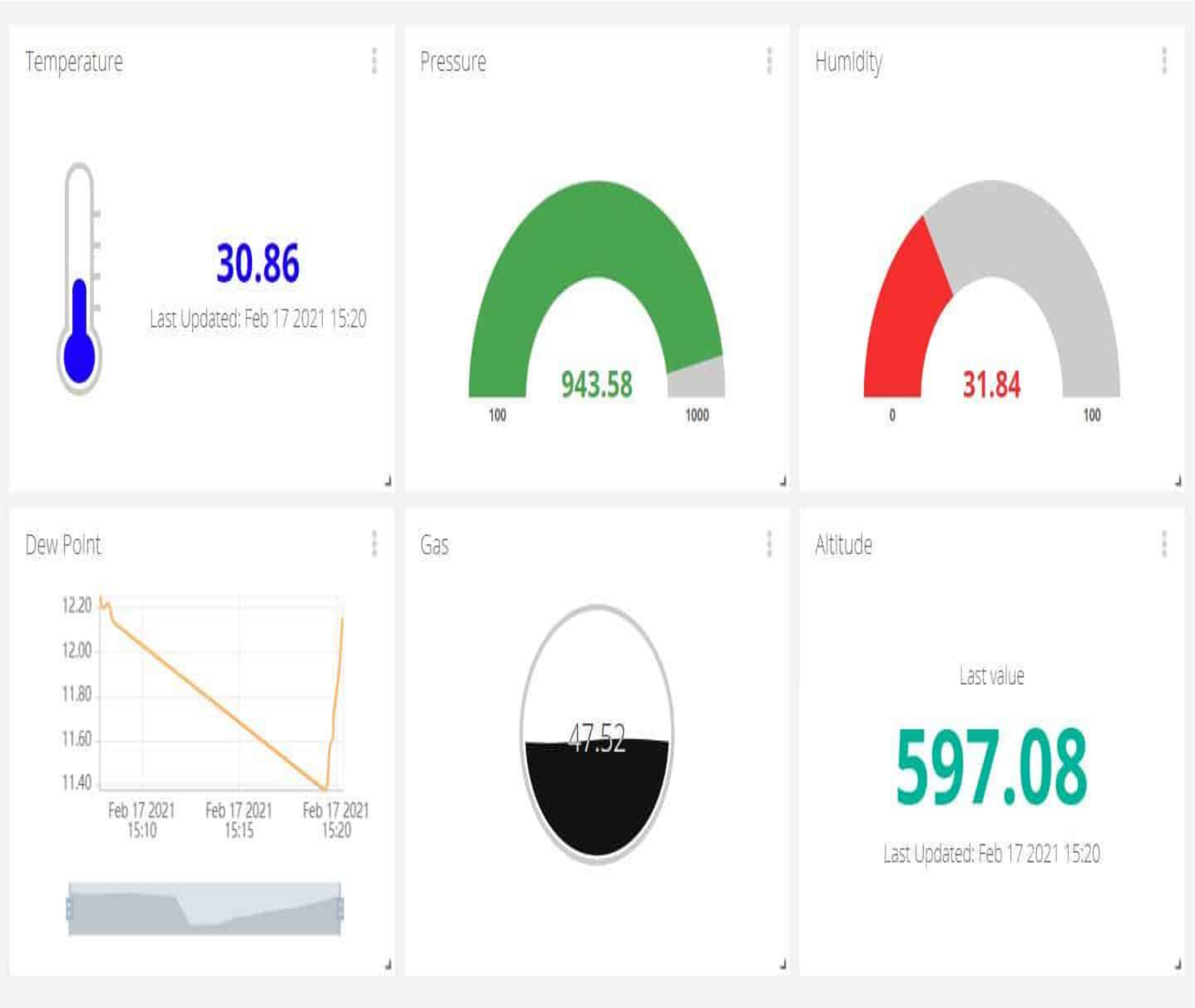


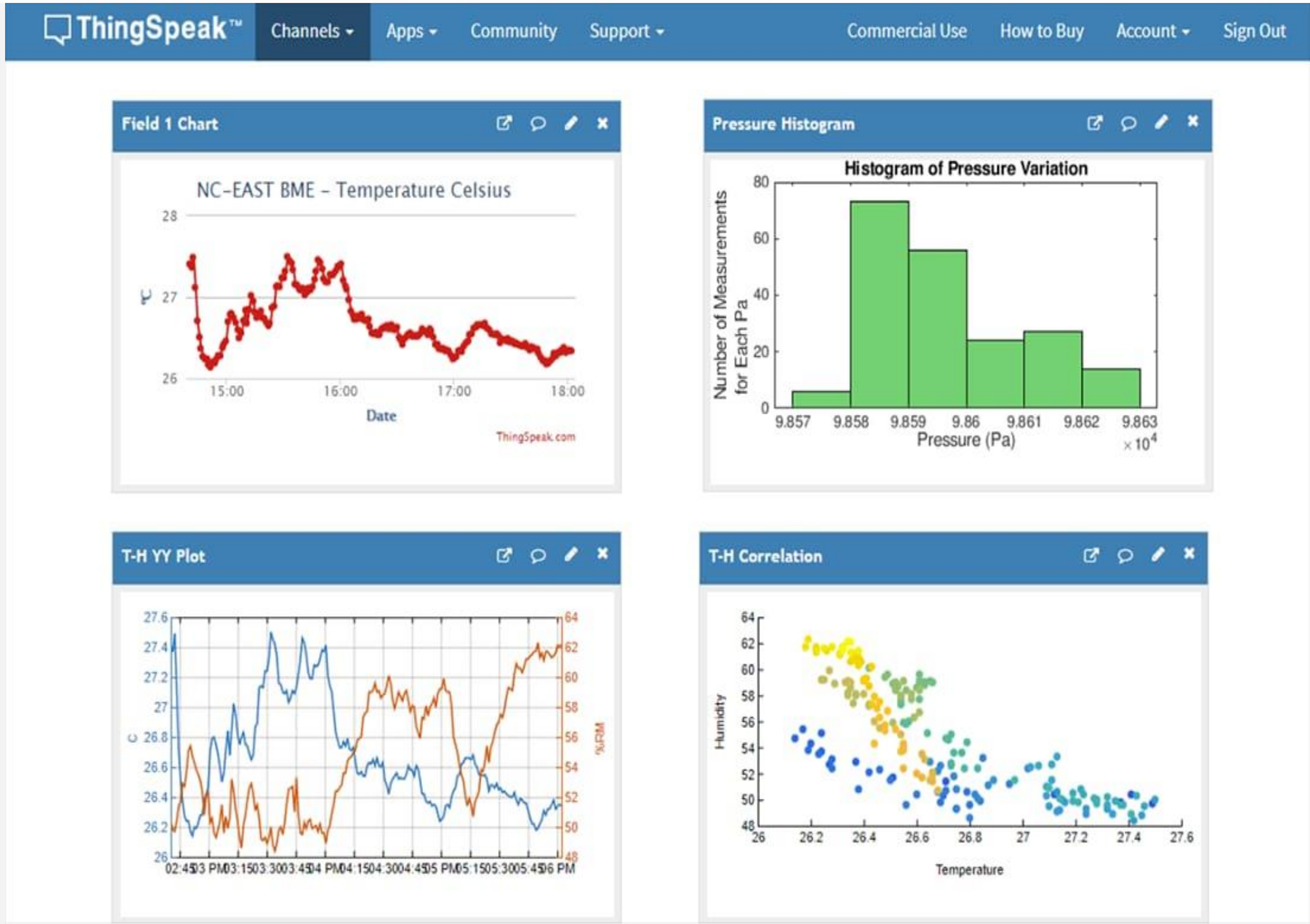
Input code

```
// Pin Definitions
#define TEMP_PIN A2
// Temperature sensor (TMP36 or similar)
#define GAS_PIN A0 // MQ Gas Sensor
#define LDR_PIN A1 // LDR
void setup() {
Serial.begin(9600);
}
void loop() {
// Temperature Sensor Reading
(TMP36)
int tempValue =
analogRead(TEMP_PIN);
float voltage = tempValue * (5.0 / 1023.0);
float temperatureC = (voltage – 0.5) * 100;
```

```
//Gas Sensor Reading
int gasValue = analogRead(GAS_PIN);
// LDR Reading
int ldrValue = analogRead(LDR_PIN);
// Print Values
Serial.println(“------ Sensor Data ------”);
Serial.print(“Temperature (C): “);
Serial.println(temperatureC);
Serial.print(“Gas Value: “);
Serial.println(gasValue);
Serial.print(“Light (LDR): “);
Serial.println(ldrValue);
Serial.println(“-------------------------”);
delay(2000); // wait 2 seconds
```

Dashboard


Temperature
30.86
Last Updated: Feb 17 2021 15:20
Pressure
943.58
100
1000
Humidity
31.84
0
100
Dew Point
12.20
12.00
11.80
11.60
11.40
Feb 17 2021 15:10
Feb 17 2021 15:15
Feb 17 2021 15:20
Gas
47.52
Altitude
Last value
597.08
Last Updated: Feb 17 2021 15:20


ThingSpeak™
Channels
Apps
Community
Support
Commercial Use
How to Buy
Account
Sign Out
Field 1 Chart
NC-EAST BME - Temperature Celsius
Date
ThingSpeak.com
Pressure Histogram
Histogram of Pressure Variation
Number of Measurements for Each Pa
Pressure (Pa)
T-H YY Plot
T-H Correlation
Humidity
Temperature

Applications & Benefits

Smart Homes

Automated temperature and humidity control for optimal comfort

Agriculture

Crop monitoring and environmental optimization for better yields

Industrial Safety

Gas detection and air quality monitoring in hazardous environments

Pollution Monitoring

Real-time air quality tracking for environmental protection

Weather Stations

Accurate environmental data collection for meteorological purposes

Smart Cities

Integrated urban environmental monitoring systems

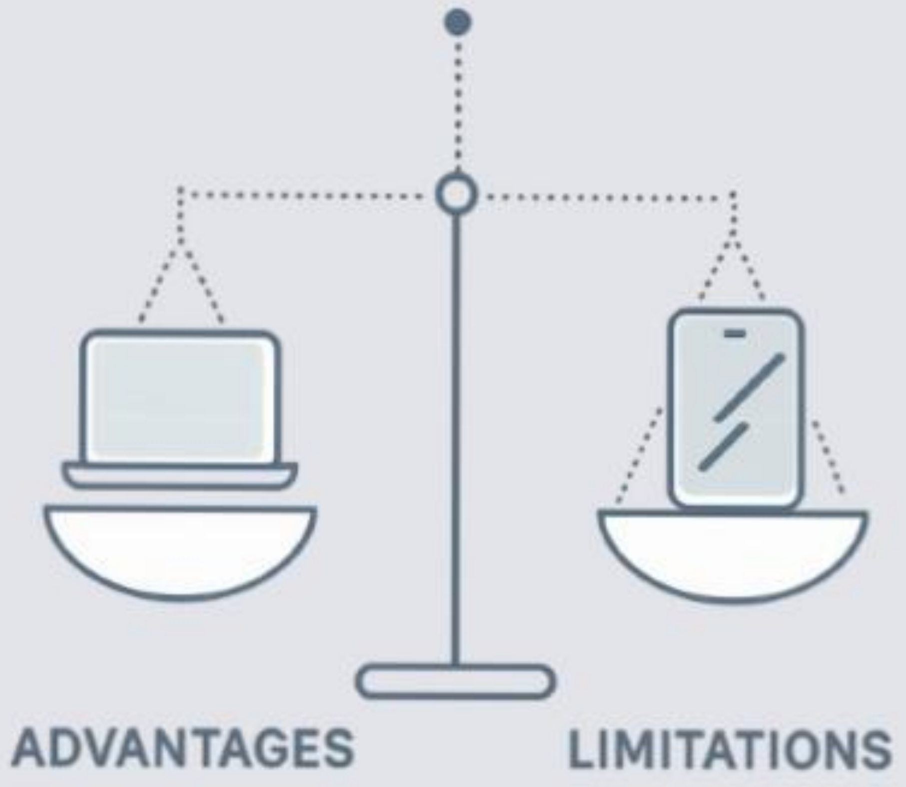
Advantages & Limitations

Advantages

- Real-time monitoring capabilities
- Remote accessibility from anywhere
- Low power consumption design
- Wireless communication eliminates cables
- Scalable and flexible architecture

Limitations

- Requires stable internet connectivity
- Sensor accuracy limitations exist
- Security risks without encryption
- Power dependency on external supply



Future Scope & Testing

AI Integration

Prediction algorithms for environmental trends

Secure Communication

Encrypted data transmission protocols

Mobile Alerts

SMS and email notifications for threshold breaches

Solar Power

Sustainable, off-grid deployment options

Smart City Systems

Integration with urban infrastructure networks

Testing Procedure

Connect sensors → Upload code → Connect to Wi-Fi → Start dashboard → Verify readings

Conclusion

The Environmental Telemetry System provides an efficient, cost-effective solution for real-time monitoring and remote data transmission. It forms the foundation for modern smart monitoring systems used across industries, agriculture, and smart cities.

24/7

Continuous Monitoring

Round-the-clock environmental tracking

100%

Wireless

Complete wireless communication

3+

Sensor Types

Temperature, humidity, gas, light

Thank you